Smart Textile Manufacturing Quality Inspection and Production Optimization Platform

SCENARIO – BASED REQUIREMENT ANALYSIS

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1.Project Vision and Stakeholders

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) is to define the functional and non-functional requirements of the **Smart Textile Manufacturing Quality Inspection and Production Optimization Platform**. This document serves as a communication medium between stakeholders, developers, testers, and project managers by providing a clear understanding of the proposed system. It describes the system objectives, scope, functional behavior, quality attributes, interfaces, constraints, and assumptions required for successful software development.

The proposed platform aims to automate textile quality inspection using Artificial Intelligence (AI), Computer Vision, Internet of Things (IoT), and Data Analytics. The system is designed to improve production quality, reduce material wastage, enhance machine utilization, and support data-driven decision-making within textile manufacturing industries.

1.2 Document Scope

This Software Requirement Specification covers all major software requirements for the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. It includes functional requirements, non-functional requirements, actors, use cases, data requirements, interface requirements, system features, assumptions, constraints, and requirement validation. The document provides a complete understanding of how the software system should operate and interact with different stakeholders.

1.3 Intended Audience

This document is intended for the following stakeholders involved in the development and implementation of the proposed system:

- Factory Owner
- Production Manager
- Quality Inspector
- Production Supervisor
- Machine Operator
- Maintenance Engineer
- System Administrator
- Software Developers

- AI/ML Engineers
- Software Testers
- Project Managers

Each stakeholder can use this document to understand the scope, functionality, and expected performance of the system.

1.4 System Overview

The Smart Textile Manufacturing Quality Inspection and Production Optimization Platform is an intelligent manufacturing management system developed to automate quality inspection and optimize production processes in textile industries. The system integrates Artificial Intelligence (AI), Computer Vision, Internet of Things (IoT), and Data Analytics to improve operational efficiency and product quality. The platform automatically detects fabric defects such as weaving faults, stitching defects, dyeing inconsistencies, stains, and surface damages during production. It also monitors machine performance, production output, downtime, and operational efficiency through real-time dashboards. By combining automated inspection with production monitoring and data analytics, the system enables textile manufacturers to reduce material wastage, minimize production costs, improve machine utilization, and support data-driven decision-making.

2. Problem Description

The textile manufacturing industry continues to rely heavily on manual inspection methods for quality assurance. Human inspectors visually examine fabrics to identify weaving faults, stitching defects, dyeing inconsistencies, stains, and other manufacturing defects. These manual inspection processes are time-consuming, inconsistent, and highly dependent on human expertise and concentration. As production volumes increase, inspectors become fatigued, resulting in reduced inspection accuracy and inconsistent quality standards. Many defects remain undetected until the final quality inspection stage, leading to material wastage, production delays, increased operational costs, customer dissatisfaction, and product rejection. In addition, production planning in many textile factories is based primarily on operator experience rather than real-time production data. Machine performance, downtime, production speed, and defect occurrence are often monitored independently, making it difficult to identify production bottlenecks or determine the root causes of recurring quality issues.

Quality reports are frequently maintained using manual records or disconnected spreadsheets, limiting the ability of management to analyze historical trends and make informed business decisions. To overcome these challenges, an integrated intelligent software platform is required that automates fabric defect detection, continuously monitors production activities, analyzes machine performance, generates quality reports, and provides production optimization recommendations based on real-time manufacturing data.

3. Scope of the System

The Smart Textile Manufacturing Quality Inspection and Production Optimization Platform is designed to improve quality assurance and production management across textile manufacturing industries by integrating Artificial Intelligence, Computer Vision, Internet of Things (IoT), and Data Analytics into a centralized software solution. The system automates the inspection of textile products using AI-powered image processing techniques capable of identifying weaving faults, stitching errors, dyeing inconsistencies, stains, and other surface defects during production. It continuously monitors machine performance, production speed, machine utilization, downtime, and production efficiency through IoT-enabled sensors and industrial equipment. The platform provides centralized dashboards for production managers, quality inspectors, supervisors, maintenance engineers, and factory owners. It stores historical production and quality data, generates automated reports, performs root-cause analysis, and recommends production optimization strategies to improve manufacturing efficiency. The proposed system supports various textile manufacturing processes including weaving, knitting, dyeing, finishing, and fabric inspection. It is scalable, allowing deployment across multiple production lines and manufacturing plants while supporting future enhancements such as predictive maintenance, cloud integration, and advanced analytics.

4. Scenario Understanding and Problem Analysis

4.1 Business Domain

The project belongs to the Textile Manufacturing Industry, specifically focusing on quality inspection, production monitoring, and manufacturing process optimization. The proposed platform integrates Artificial Intelligence (AI), Computer Vision, Internet of Things (IoT), and Data Analytics to automate quality inspection, monitor manufacturing operations, and optimize production performance. It supports the digital transformation of traditional textile manufacturing by replacing manual inspection methods with intelligent automated solutions.

4.2 Existing Challenges

The current textile manufacturing process faces several operational and quality-related challenges, including:

- Manual inspection requires significant time and skilled labor.
- Human inspection accuracy decreases due to fatigue and repetitive tasks.
- Fabric defects are often detected only during final quality inspection.
- Material wastage increases because defective products continue through multiple production stages.
- Machine performance and production efficiency are not continuously monitored.
- Production scheduling depends heavily on operator experience rather than data-driven decision-making.
- Root causes of recurring defects are difficult to identify due to the absence of integrated production data.
- Manual report preparation limits historical analysis and decision-making.

4.3 Product Backlog Summary

Stakeholder	Role and Responsibility
Factory Owner	Reviews production reports, monitors business performance, and makes strategic decisions.
Production Manager	Supervises production planning, monitors machine utilization, and improves manufacturing efficiency.
Quality Inspector	Verifies AI-generated inspection results and ensures product quality standards are maintained.
Production Supervisor	Monitors production activities and responds to quality alerts.
Machine Operator	Operates textile manufacturing equipment and monitors machine conditions.
Maintenance Engineer	Performs preventive and corrective maintenance to minimize machine downtime.
System Administrator	Manages user accounts, system security, database configuration, and software maintenance.
Customers	Receive high-quality textile products with minimal defects and timely delivery.

4.4 System Objectives

The objectives of the proposed system are:

- Automate fabric defect detection using Artificial Intelligence and Computer Vision.
- Improve the accuracy and consistency of textile quality inspection.
- Monitor production activities and machine performance in real time.
- Reduce material wastage by detecting defects at an early stage.
- Classify defects according to type, severity, and location.
- Perform root-cause analysis using historical production and quality data.
- Generate automatic alerts for high defect rates and machine failures.
- Optimize machine utilization and production scheduling using data-driven recommendations.
- Generate centralized production and quality reports.
- Improve overall manufacturing efficiency and customer satisfaction.

4.5 Proposed Solution

The proposed Smart Textile Manufacturing Quality Inspection and Production Optimization Platform integrates AI-powered image processing, IoT-based production monitoring, centralized data management, and intelligent analytics into a single software solution. The platform automatically detects and classifies fabric defects, continuously monitors machine performance, tracks production efficiency, generates intelligent alerts, performs root-cause analysis, and provides optimization recommendations through interactive dashboards and analytical reports. By replacing manual inspection with automated quality assurance and real-time production monitoring, the system helps textile manufacturers reduce operational costs, improve product quality, minimize production delays, and enhance decision-making based on accurate manufacturing data.

5. Requirements

5.1 Introduction

Functional requirements define the services, features, and operations that the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform must perform. These requirements describe the expected behavior of the system and the interactions between users and the software. Each requirement is assigned a unique identifier for easy reference during design, implementation, testing, and maintenance.

5.2 Functional Requirements

FR-01: User Registration

The system shall allow the System Administrator to create user accounts for authorized employees by entering user details such as name, email, username, password, and assigned role.

FR-02: User Authentication

The system shall authenticate registered users using valid login credentials before granting access to the platform.

FR-03: Role-Based Access Control

The system shall provide role-based access permissions for different users, including Factory Owner, Production Manager, Quality Inspector, Production Supervisor, Machine Operator, Maintenance Engineer, and System Administrator.

FR-04: Fabric Image Acquisition

The system shall capture high-resolution fabric images from industrial cameras installed along the production line for automated inspection.

FR-05: Automated Fabric Defect Detection

The system shall automatically detect fabric defects such as weaving faults, stitching defects, dyeing inconsistencies, stains, holes, and surface damages using Artificial Intelligence and Computer Vision algorithms.

FR-06: Defect Classification

The system shall classify detected defects according to their type, severity level, and location on the fabric.

FR-07: Defect Data Storage

The system shall store all detected defect information, including images, defect category, severity, inspection time, production line, and batch details, in the centralized database.

FR-08: Production Monitoring

The system shall continuously monitor machine performance, production output, machine utilization, operational status, and downtime in real time.

FR-09: Production Dashboard

The system shall display live production statistics, machine status, production efficiency, defect count, and downtime information through an interactive dashboard.

FR-10: Root Cause Analysis

The system shall analyze historical production data to identify recurring defect patterns and determine possible root causes associated with machines, operators, shifts, or raw material batches.

FR-11: Alert Generation

The system shall automatically generate notifications whenever the defect rate exceeds predefined quality thresholds.

FR-12: Machine Failure Notification

The system shall notify maintenance engineers whenever abnormal machine conditions or equipment failures are detected.

FR-13: Production Report Generation

The system shall generate daily, weekly, and monthly production reports containing production statistics, machine utilization, downtime, and overall manufacturing performance.

FR-14: Quality Report Generation

The system shall generate detailed quality reports containing defect statistics, defect classifications, inspection history, and quality trends.

FR-15: Production Optimization Recommendations

The system shall recommend optimized machine speeds and production schedules based on historical production data and current manufacturing conditions.

FR-16: Search and Filter Records
The system shall allow authorized users to search production records using parameters such as production date, machine ID, batch number, defect type, operator, and shift.

FR-17: Report Export
The system shall allow authorized users to export production and quality reports in PDF and Microsoft Excel formats.

FR-18: Audit Log Management
The system shall maintain a complete audit log of user activities, system events, and administrative actions for security and traceability purposes.

5.3 Functional Requirement Summary

Requirement ID	Functional Module
FR-01	User Registration
FR-02	User Authentication
FR-03	Role-Based Access Control
FR-04	Fabric Image Acquisition
FR-05	Automated Fabric Defect Detection
FR-06	Defect Classification
FR-07	Defect Data Storage
FR-08	Production Monitoring
FR-09	Production Dashboard
FR-10	Root Cause Analysis
FR-11	Alert Generation
FR-12	Machine Failure Notification
FR-13	Production Report Generation
FR-14	Quality Report Generation
FR-15	Production Optimization Recommendations

6.Non-Functional Requirements

6.1 Introduction

Non-functional requirements define the quality attributes and performance characteristics of the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. These requirements ensure that the system is secure, reliable, scalable, efficient, and easy to maintain.

6.2 Performance Requirements

- The system shall process and analyze fabric images within 3 seconds of image capture.
- The production dashboard shall refresh real-time data every 5 seconds.
- The system shall support at least 200 concurrent users without significant performance degradation.
- Production reports shall be generated within 10 seconds.

6.3 Security Requirements

- All users shall authenticate using secure login credentials.
- Passwords shall be securely stored using encryption.
- Role-based access control shall restrict unauthorized access to sensitive information.
- User sessions shall automatically expire after a period of inactivity.
- All user activities shall be recorded in the audit log.

6.4 Reliability Requirements

- The system shall maintain 99.9% availability during production hours.
- Automatic database backup shall be performed daily.
- Data loss shall be minimized in the event of system failure.
- The system shall recover automatically after unexpected failures whenever possible.

6.5 Usability Requirements

- The user interface shall be simple, intuitive, and easy to navigate.
- Dashboard information shall be displayed using charts, tables, and graphical indicators.
- The system shall provide meaningful error messages and user guidance.
- Users shall be able to perform common operations with minimal training.

6.6 Scalability Requirements

- The platform shall support multiple textile manufacturing plants.
- Additional production lines and machines shall be integrated without major software modifications.
- The system architecture shall support future AI model upgrades and cloud integration.

6.7 Maintainability Requirements

- The software shall be modular to simplify future enhancements.
- System documentation shall be maintained for all software modules.
- Software updates shall not interrupt production operations.
- Bugs shall be resolved through version-controlled software releases.

6.8 Availability Requirements

- The platform shall be accessible 24 hours a day, 7 days a week.
- Scheduled maintenance shall be performed during non-production hours.
- Automatic failover mechanisms shall ensure uninterrupted monitoring services.

6.9 Non-Functional Requirement Summary

Requirement Category	**Description**
Performance	Fast image processing and real-time monitoring
Security	Secure authentication, encryption, and access control
Reliability	High availability and automatic recovery
Usability	User-friendly interface and easy navigation
Scalability	Support for multiple factories and production lines
Maintainability	Easy software maintenance and future upgrades

7. Actor Identification and Use Case Modelling

7.1 Introduction

Actor Identification and Use Case Modelling describe the interactions between users and the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. An actor represents a person or external system that communicates with the application to perform specific operations. Use cases represent the major functionalities offered by the system to achieve the business objectives of automated quality inspection and production optimization.

7.2 Actor Identification

The Smart Textile Manufacturing Quality Inspection and Production Optimization Platform involves multiple stakeholders who interact with the system during daily manufacturing operations. Each actor has specific responsibilities based on their role within the organization.

Actor	**Description**
Factory Owner	Reviews production reports, quality reports, and overall manufacturing performance.
Production Manager	Monitors production activities, machine utilization, and production schedules.
Quality Inspector	Reviews AI-generated inspection results and validates detected defects.
Production Supervisor	Monitors production operations and responds to system alerts.
Maintenance Engineer	Receives machine failure notifications and performs maintenance activities.
Machine Operator	Operates textile manufacturing equipment and monitors machine conditions.
System Administrator	Manages users, system configuration, security, and software maintenance.

AI Defect Detection Engine	Automatically analyzes fabric images and detects defects.
Industrial Camera	Captures fabric images during the manufacturing process.
IoT Sensors	Collect production and machine performance data.

7.3 Use Cases

The major use cases supported by the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform are summarized below.

Use Case ID	Use Case Name	Primary Actor
UC-01	User Login	All Users
UC-02	Manage User Accounts	System Administrator
UC-03	Capture Fabric Images	Industrial Camera
UC-04	Detect Fabric Defects	AI Defect Detection Engine
UC-05	Classify Defects	Quality Inspector
UC-06	Monitor Production Dashboard	Production Manager
UC-07	Monitor Machine Performance	Production Supervisor
UC-08	Perform Root Cause Analysis	Production Manager
UC-09	Generate Alerts	System
UC-10	Receive Machine Failure Notifications	Maintenance Engineer
UC-11	Generate Production Reports	Factory Owner
UC-12	Generate Quality Reports	Quality Inspector
UC-13	View Production Optimization Recommendations	Production Manager

7.4 Chapter Summary

This chapter identified all the actors involved in the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform and summarized the major use cases performed by each actor. The Use Case Diagram visually represents these interactions and provides a clear understanding of the system's functional behavior.

8. Assumptions, Constraints, Requirement Validation and Conclusion

8.1 Assumptions

The Smart Textile Manufacturing Quality Inspection and Production Optimization Platform is developed based on several assumptions regarding the operating environment and available resources. It is assumed that all textile manufacturing machines are equipped with industrial cameras and IoT sensors capable of capturing production and machine data accurately. The manufacturing unit is expected to have a reliable network infrastructure to support continuous communication between hardware devices and the software platform. It is also assumed that authorized users receive adequate training before operating the system and that the organization maintains a centralized database for storing production and inspection records.

8.2 Constraints

The development and implementation of the proposed system are influenced by several constraints. From a technical perspective, the platform must remain compatible with existing manufacturing equipment and industrial cameras. Operationally, the software should function without interrupting normal production activities. Economic constraints include the budget available for AI implementation, hardware procurement, and system deployment. The system must comply with organizational security policies and applicable data protection regulations. In addition, the project must be completed within the specified development schedule while maintaining the required uality standards.

8.3 Requirement Validation

The Agile Sprint Planning process provides a structured approach to developing the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. By defining Epics, User Stories, Acceptance Criteria, Product Backlog, Sprint Backlog, Story Point Estimation, Sprint Goals, and Expected Deliverables, the development team can effectively manage project requirements, prioritize tasks, and deliver high-quality software incrementally.

8.4 Conclusion

The Smart Textile Manufacturing Quality Inspection and Production Optimization Platform provides an integrated solution for modern textile manufacturing by combining Artificial Intelligence, Computer Vision, Internet of Things, and data analytics. The proposed system automates fabric defect detection, improves production monitoring, enhances quality control, and supports data-driven decision-making through centralized dashboards and analytical reports. The Software Requirement Specification presented in this document establishes a clear and comprehensive foundation for system design, development, testing, and deployment while addressing the challenges identified in the problem statement. The implementation of this platform is expected to improve manufacturing efficiency, reduce operational costs, minimize material wastage, and enhance the overall quality of textile products